G - Balanced Subarrays

分值：625 分

Problem Statement

You are given a sequence A of N positive integers.
Among the $\frac{N(N+1)}{2}$ non-empty (contiguous) subarrays of A, we call a sequence X a balanced sequence if and only if it satisfies the following condition:

给定一个长度为 N 的正整数序列 A。在 A 的所有 $\frac{N(N+1)}{2}$ 个非空（连续）子数组中，我们称序列 X 为平衡序列当且仅当它满足以下条件：

- Every integer appearing in X appears the same number of times in X.
- 所有在 X 中出现过的整数，在 X 中的出现次数均相等。

For each $k = 1, 2, \ldots, K$, find the following two values:

对于每个 $k = 1, 2, \ldots, K$，请求出以下两个值：

- The number of balanced sequences in which each integer appears exactly B_k times.
- The number of balanced sequences in which exactly B_k distinct integers appear.
- 每个整数恰好出现 B_k 次的平衡序列的数量。
- 恰好包含 B_k 个不同整数的平衡序列的数量。

Count two subarrays separately if they are taken from different positions in A, even if they are identical as sequences.

只要两个子数组在 A 中的位置不同，即使它们作为序列完全相同，也需要单独计数。

Solve T test cases per input.

每个输入包含 T 组测试用例，你需要依次处理。

Constraints

- $1 \leq T \leq 2 \times 10^5$
- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq \min(N, 10)$
- $1 \leq A_i \leq N$
- $1 \leq B_k \leq N$
- $B_1, B_2, \ldots, B_K$ are pairwise distinct.
- The sum of N over all test cases is at most 2×10^5.
- All input values are integers.
- $1 \leq T \leq 2 \times 10^5$
- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq \min(N, 10)$
- $1 \leq A_i \leq N$
- $1 \leq B_k \leq N$
- $B_1, B_2, \ldots, B_K$ 两两不同。
- 所有测试用例的 N 之和不超过 2×10^5。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$$
\begin{array}{l}
T \\
\mathrm{case}_1 \\
\mathrm{case}_2 \\
\vdots \\
\mathrm{case}_T
\end{array}
$$

Each case is given in the following format:

每组测试用例按照以下格式给出：

N K
A_1 A_2 $\ldots$ A_N
B_1 B_2 $\ldots$ B_K

Output

Output the answers in the following format:

按照以下格式输出答案：

```
$\mathrm{case}_1$
$\mathrm{case}_2$
$\vdots$
$\mathrm{case}_T$
```


```
$\mathrm{case}_1$
$\mathrm{case}_2$
$\vdots$
$\mathrm{case}_T$
```

For each case, letting $c_{k,1}$ be the number of balanced sequences in which each integer appears exactly B_k times, and $c_{k,2}$ be the number of balanced sequences in which exactly B_k distinct integers appear, output in the following format:

对于每组测试用例，设 $c_{k,1}$ 为每个整数恰好出现 B_k 次的平衡序列的数量，$c_{k,2}$ 为恰好包含 B_k 个不同整数的平衡序列的数量，请按照以下格式输出：

```
$c_{1,1}$ $c_{1,2}$
$c_{2,1}$ $c_{2,2}$
$\vdots$
$c_{K,1}$ $c_{K,2}$
```


```
$c_{1,1}$ $c_{1,2}$
$c_{2,1}$ $c_{2,2}$
$\vdots$
$c_{K,1}$ $c_{K,2}$
```

Sample Input 1

```
3
4 2
1 2 1 2
2 1
1 1
1
1
7 7
1 5 5 1 5 1 2
1 2 3 4 5 6 7
```

Sample Output 1

```
1 4
7 4
1 1
13 8
3 8
1 1
0 0
0 0
0 0
0 0
```

For the first test case, the pairs (l, r) such that the subarray of A from the l-th through the r-th element is a balanced sequence are $(1,1), (1,2), (1,4), (2,2), (2,3), (3,3), (3,4), (4,4)$, giving eight instances in total.
Among these, the ones in which each integer appears exactly twice are $(1,4)$, giving one instance, and the ones in which exactly two distinct integers appear are $(1,2), (1,4), (2,3), (3,4)$, giving four instances.
Thus, for $k = 1$, output `1 4`.
Also, the ones in which each integer appears exactly once are $(1,1), (1,2), (2,2), (2,3), (3,3), (3,4), (4,4)$, giving seven instances, and the ones in which exactly one distinct integer appears are $(1,1), (2,2), (3,3), (4,4)$, giving four instances.
Thus, for $k = 2$, output `7 4`.

对于第一组测试用例，所有满足“A 中从第 l 个元素到第 r 个元素的子数组是平衡序列”的数对 (l, r) 为 $(1,1),(1,2),(1,4),(2,2),(2,3),(3,3),(3,4),(4,4)$，总共有 8 个。其中，每个整数恰好出现两次的平衡序列为 $(1,4)$，共 1 个；恰好包含两个不同整数的平衡序列为 $(1,2),(1,4),(2,3),(3,4)$，共 4 个。因此对于 $k=1$，输出 `1 4`。此外，每个整数恰好出现一次的平衡序列为 $(1,1),(1,2),(2,2),(2,3),(3,3),(3,4),(4,4)$，共 7 个；恰好包含一个不同整数的平衡序列为 $(1,1),(2,2),(3,3),(4,4)$，共 4 个。因此对于 $k=2$，输出 `7 4`。